SPACE FACULTY

Let’s learn about **satellites**!

Space mission and **satellite** design course

Mission Design - Let's design a satellite mission

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Let’s Design a Cubesat Mission!

You will use the concepts and tools from our previous sessions to design a complete space mission from scratch.

We will begin by identifying a real operational need. Your task will then be to define a mission concept that addresses it, objectives, orbit, payload, platform choices, and a realistic operations approach.

This exercise will help you consolidate what you have learned and practice applying it in a practical, real-world context.

Let’s see what mission ideas you can come up with!

Useful Websites

The **Nanosats Database** is the world's largest database of nanosatellites, including CubeSat constellations, companies, technologies, and missions.

At the following web address, you will find a list of CubeSat companies offering missions, platforms, components, software, and services for or with nanosatellites.

https://www.nanosats.eu/companies

Satsearch is a global marketplace for the space industry, connecting engineers with suppliers worldwide.

https://satsearch.co/

SpaceX rideshare program: find all the information you need to make a reservation online, everything from plate configuration to technical specifications to licensing information.

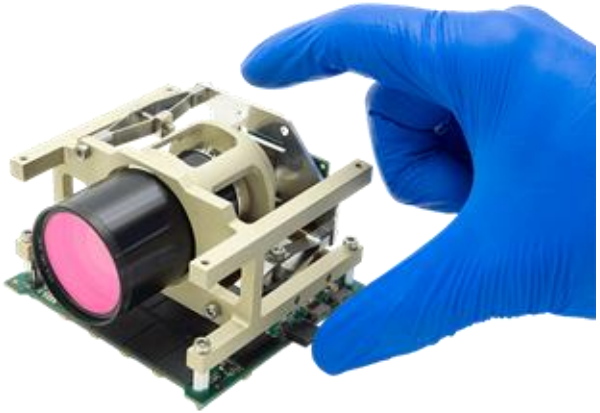
https://www.spacex.com/rideshare/

Mission Explanation

SpaceYZ is a startup that has just closed its first round of investment, securing $5 million. Now, the investors are urging the company to conduct an IOD (In-Orbit Demonstration) of their product.

SpaceYZ has designed a new imaging camera that incorporates AI capabilities:

- 2U form factor (camera dimension)
- Operating voltage 5V
- Power consumption 10W
- Mass 1.3kg



Mission Explanation

The CTO of SpaceYZ is contacting your company to handle the management of the IOD mission and is open to considering various options and quotations.

This is their first time conducting an IOD, and they are seeking a reputable company to oversee the entire mission, including design, assembly, integration, testing, and ground operations. They understand the importance of getting it right the first time to impress the investors.

What they expect you to provide?

- Platform selection (CubeSat and components).
- Mission dedicated personnel.
- Facility and testing.
- Launch opportunity.
- Ground station operations.

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What You Must Deliver

They want a short proposal in PDF format (2–3 pages).

Your document must include:

CubeSat platform proposal
- Size and basic specifications
- Any additional components needed
- Why this platform is a good fit (reliability, compatibility, cost)

Dedicated team
- Roles you will allocate (PMs, engineers, technicians, etc.)
- Why this team is credible (skills and experience)

Facilities and testing approach
- How you will assemble, integrate, and test the CubeSat
- How you will ensure readiness for launch

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Additional Required Content

Your PDF should also cover:

Launch strategy
- Potential launch providers and options
- Expected timeline
- Contingency plan for delays

Ground operations concept
- Ground station location(s)
- Frequency bands and communications approach
- Data transfer method
- Redundancy / backup plan

Project plan and commercial info
- Timeline, milestones, and deliverables
- Payment plan (upfront + milestone payments)
- Cost breakdown (hardware, personnel, testing, launch, operations)

Optional
- Conceptual drawings / renderings of the CubeSat solution

Mission Objectives Structure

Include the following information at the beginning of the document:

Mission Name:	*Give a name to the CubeSat mission*
Mission Objective 1 (O1):	*Define the first objective of the mission*
Mission Objective 2 (O2):	*Define the second objective of the mission*
Mission Objective 3 (O3):	*Define the third objective of the mission*

The first objective of the mission represents the best outcome from the In-Orbit testing. The second objective represents valuable information that is still obtained even if the first objective fails. The third objective represents something of lesser importance than the second but still holds value. If all objectives are successful, we achieve mission success at 120% (100% for the first objective, 110% for the second, and 120% for the third).

Example of Objectives (UHF Radio IOD)

Let's have a look at an example for a mission that involves the testing of a new UHF radio with antenna:

Objective 1 (O1) – Primary
- Demonstrate successful TX/RX using the new radio and antenna
- Verify good performance in different CubeSat orientations and distances
- Check robustness against interference

Objective 2 (O2) – Secondary
- Collect reliable communication data even if performance is not optimal
- Evaluate resilience through metrics like signal strength and error rates

Objective 3 (O3) – Additional
- Gather extra information (e.g., antenna deployment success)
- Optional tests: power consumption, radiation pattern, compatibility
- Useful for next design iteration

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